Team Number: 89

M2 — Automatic Category Suggestions for Wikimedia Commons

Problem Statement 3 · Progress report

1. Current approach and changes since M1

The M1 architecture is fully implemented and unchanged: a frozen CLIP ViT-B/32 encoder; four scorers (k-NN voting over categorised images, zero-shot category-name prototypes, a text-initialised multi-label head trained with asymmetric loss, and a metadata scorer using title/description embedding, word overlap and year agreement); logistic-regression fusion over the union of candidates; hierarchy post-processing; live index updates from editor feedback and new categories. Working end to end: Commons crawler, training, evaluation harness, command-line predictor, and a web demo with evidence images and accept/reject feedback. Changes since M1, each driven by an experiment below:

- **Realistic validation.** The vocabulary is now built only from data that precedes the validation period; validation categories are added after fusion is fitted. Before, every validation label was “known”, so fusion was tuned on an easy set (val F1@5 0.307 vs test 0.172). Now val 0.167 vs test 0.153.
- **Hierarchy strength tuned on validation** (1.0 = off) instead of fixed by hand; on realistic validation it selects 1.0, so suppression no longer costs accuracy.
- **Less biased sampling and a bias audit:** region-balanced seeds (8 countries per continent), optional uniformly random Commons files, and accuracy reported per region and topic group.
- **Engineering:** exact NumPy search on macOS (FAISS and PyTorch can deadlock there); a crawler fix for an API edge case.

2. Experiments and results

Data. Commons files crawled through the MediaWiki API; labels are editor-assigned categories after removing hidden and maintenance categories; temporal split by upload time (old → train, newest → test) with near-duplicate removal (perceptual hash + embedding cosine ≥ 0.95). Run 1: 1,279 files from 107 topical seeds. Run 2: 3,416 files, more per category. **Run 3: 6,303 files** (4,429 train / 937 val / 937 test) from 155 seeds (107 topical + 48 region-balanced) plus 1,500 uniformly random files, with realistic validation; vocabulary 24,486 categories, 466 in the supervised head; k-NN tuned to K = 50, τ = 0.1.

Table 1. Runs compared (test, image + metadata).

metric	Run 1	Run 2	Run 3
test images	190	510	937
P@1	0.284	0.339	**0.350**
F1@1	0.148	0.199	**0.237**
F1@5	0.119	**0.160**	0.153
R@10	0.238	0.310	**0.330**
F1@5, image only	0.089	**0.112**	0.084
label coverage	0.335	0.377	0.362
val − test F1@5 gap	–	0.135	**0.014**

Table 2. Run 3, fusion (sample-wise).

input	k	P@k	R@k	F1@k	hier. P
image + metadata	1	0.350	0.203	0.237	0.425
image + metadata	5	0.116	0.304	0.153	0.240
image + metadata	10	0.066	0.330	0.103	0.174
image only	1	0.175	0.093	0.112	0.232
image only	5	0.064	0.163	0.084	0.119

Table 3. Run 3 ablation, image + metadata.

ranking	F1@5	R@10
fusion (all scorers)	**0.153**	**0.330**
metadata lexical only	0.100	0.203
supervised head only	0.070	0.173
k-NN only	0.069	0.156
metadata text only	0.052	0.170
zero-shot only	0.005	0.020

Table 4. Run 3 recall@5 by frequency in the index.

bucket	categories	labels	macro R
unseen (0)	753	2,412	0.004
tail (1–4)	71	318	0.214
torso (5–19)	24	217	0.434
head (20+)	72	380	0.830

Table 5. Run 3 bias audit (groups with ≥ 10 test images).

group	images	F1@5	R@10	F1@5 image only
Objects	318	0.192	0.408	0.140
Scenes	109	0.232	0.422	0.162
Events	20	0.233	0.434	0.200
Abstract concepts	127	0.050	0.085	0.029
Europe	42	0.278	0.735	0.037
Africa	71	0.204	0.597	0.009
South America	48	0.103	0.239	0.009
Asia	47	0.087	0.162	0.038
N. America & Caribbean	43	0.032	0.065	0.012

Table 6. Run 3 robustness, F1@5 (share flagged low-quality).

degradation	flagged	+ metadata	image only
none	–	0.153	0.084
Gaussian blur r=2	1.00	0.155	0.079
Gaussian blur r=5	1.00	0.147	0.054
JPEG quality 10	0.35	0.151	0.072
downscale ×0.15	1.00	0.151	0.066

Efficiency (team's Mac, no server GPU): end-to-end per image p50 19.6 ms, p95 114.8 ms; scoring and ranking 0.89 ms/image; index of 5,366 images and 24,486 categories uses 58.3 MB in memory.

Findings. (1) **Realistic validation fixed model selection:** the val–test gap fell from 0.135 to 0.014 F1@5, and the tuned hierarchy rule no longer hurts. (2) **More data improves the top of the list:** P@1 rose from 0.284 to 0.350 and R@10 from 0.238 to 0.330; F1@5 is flat on Run 3's harder test set, whose region-balanced seeds add many place-specific categories. (3) **Examples per category drive accuracy:** recall@5 is 0.830 with ≥ 20 examples but 0.004 for unseen categories, which hold 72% of test labels; label coverage (0.36) is the main ceiling. (4) **Fusion** beats the best single scorer by 53% F1@5. (5) **Metadata is essential for places and regions:** with metadata, Africa reaches F1@5 0.204 (image only 0.009); under heavy blur, F1@5 with metadata falls only 4% vs 36% image only. (6) **Bias audit:** accuracy varies widely by region (Europe 0.278, North America 0.032) and abstract concepts are weakest (0.050); groups are small (42–318 images), so differences are indicative. (7) **Zero-shot name matching alone collapses** over 24k names (F1@5 0.005), and random files fell almost entirely into training because they span Commons' whole history (≤ 8 of 937 test images).

3. Progress on testing

- Evaluation harness implements every metric planned in M1 (P/R/F1@k, mAP, hierarchy-aware precision, frequency buckets, category types, resolution bands, degradations, latency, memory, per-scorer ablation) plus the per-region bias audit.
- 61 automated tests: category filtering, API edge cases, random sampling, splits, metrics, index search (FAISS and NumPy), hierarchy rules, head, fusion, an offline end-to-end run on synthetic data, and encoder API checks on tiny CLIP and SigLIP models.
- Continual learning verified: categories added at run time and accepted feedback are suggested immediately without retraining; rejected labels act as negative votes.

4. Plan for the remaining time

- **Unbiased test set of real new uploads:** sample the most recent Commons uploads directly, so the test set matches the target "new images" distribution.
- **Encoder comparison:** SigLIP-base vs CLIP ViT-B/32 on the same split, aimed at the weak zero-shot scorer; report the accuracy/latency trade-off.
- **Raise label coverage:** add category names without images (seed subtrees, siblings of known categories) for the zero-shot and lexical scorers.
- **Reviewed score:** manually check a sample of "wrong" suggestions to estimate how many are correct but unlabelled.
- **Deliverables:** demo video (web UI with evidence and live feedback), final report, slides, externally hosted index.